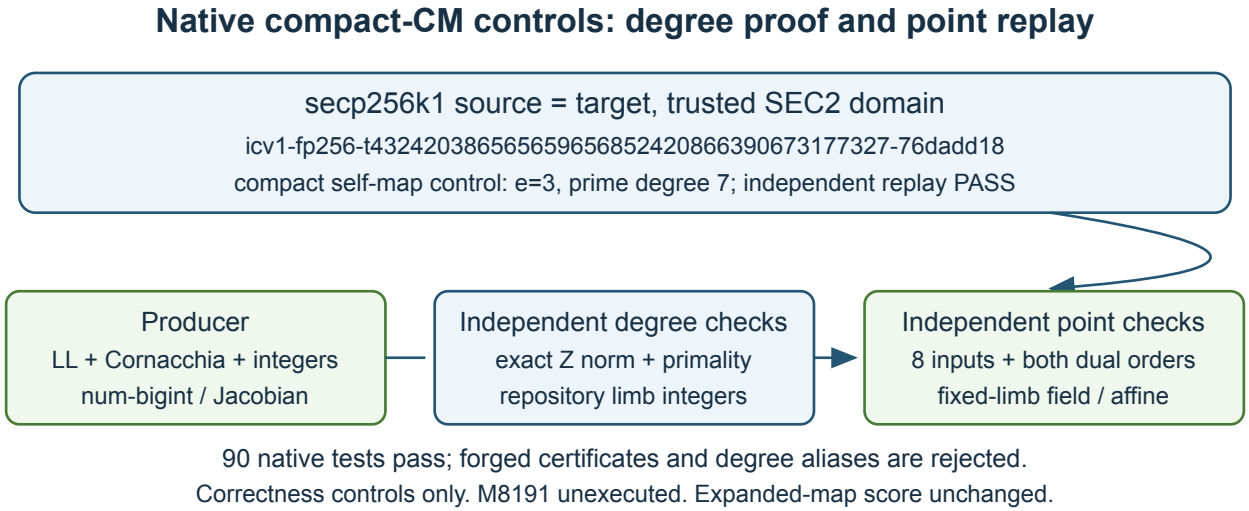
Native compact-CM certificates

Implementation and controls · 16 October 2026 UTC

The installed `isogeny` binary now exposes compact-CM construction, fresh independent verification and named geometric/subgroup evaluation. This round validates controls; the proposed degree 2^8191−1 has not been executed. Compact known-endomorphism self-maps form a separate category from the main expanded-map challenge.



What ran

The complete dedicated-tool suite passed 90 tests with zero failures: 3 field checks, 81 main-binary tests, 4 registry-model tests and two CLI integrations. Fixed prime-degree controls are 7, 31 and 127. The installed CLI is invoked from outside the checkout. Its positive construction/replay, malformed certificates, resource receipts and point images are retained under `native-controls-final-retry1/`; the final log and exit-code file bind the validation outcome. Root release-library validation separately passed 4148 tests, with 117 ignored and zero failed, at unchanged root source.

The producer uses num-bigint, Mersenne reduction and Jacobian scalar multiplication. The independent verifier uses repository limb integers with general modular reduction, independent Cornacchia replay, dense exact integer identity expansion, and fixed-limb affine point arithmetic. It checks eight fixed inputs, subgroup action, group compatibility and both dual compositions. The public constructor/verifier use native watchdogs and sampled RSS protection; no performance gain is claimed.

Proof and rejection boundaries

Evaluated supplement `IDC1hf56b3716b0a7eb8b` is accompanied by exact integer coefficient checks. A deliberately wrong integer coefficient invisible modulo the supplement's field is rejected. Wrong beta/lambda pairing is rejected. Adding the subgroup order to a coefficient preserves rational-point images while changing the geometric norm, so its original degree certificate fails. Prime exponent 11 with composite M11 and composite exponent 9 are distinct negative controls. Earlier compile/test logs are preserved.

The exact SEC2 secp256k1 tuple is checked against the pinned registry; p,n primality is explicitly trusted from SEC2, rather than self-contained here. The new degree's primality has its own independent Lucas–Lehmer replay. The reviewed dual-norm theorem establishes geometric degree from unreduced integers; point agreement cannot do so. Frobenius fixtures outside this canonical schema remain theorem-level controls.

Provenance and next gate

Source hashes, logs and the editable SVG are beside this report. Mathematical review and implementation handoff are in PR2221 implementation publication is PR1976. The exact identity and proof basis are in `docs/identity-certificates/cm-eisenstein-norm.{json,md}`. The next scientific gate requires a clean executable, immutable custody and a frozen target protocol. No worldwide record, new expanded-map value or comparative timing result is established by these controls.